

Michael Crawshaw

Ph.D. Student

George Mason University

Research Interests: Optimization for machine learning, Distributed optimization

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🌐 [Personal Website](#)

🔗 [Google Scholar](#)

EDUCATION

George Mason University

Ph.D. in Computer Science

2019 - Present

M.S. in Computer Science

2019 - 2022

Advisor: Mingrui Liu

The Ohio State University

2015 - 2019

B.S. in Mathematics and Computer Science, Honors

PUBLICATIONS

Constant Stepsize Local GD for Logistic Regression: Acceleration by Instability

Michael Crawshaw, Blake Woodworth, Mingrui Liu

International Conference on Machine Learning, 2025.

Local Steps Speed up Local GD for Heterogeneous Distributed Logistic Regression

Michael Crawshaw, Blake Woodworth, Mingrui Liu

International Conference on Learning Representations, 2025.

Complexity Lower Bounds of Adaptive Gradient Algorithms for Non-convex Stochastic Optimization under Relaxed Smoothness

Michael Crawshaw, Mingrui Liu

International Conference on Learning Representations, 2025.

Federated Learning under Periodic Client Participation and Heterogeneous Data: A New Communication-Efficient Algorithm and Analysis

Michael Crawshaw, Mingrui Liu

Conference on Neural Information Processing Systems, 2024.

Provable Benefits of Local Steps in Heterogeneous Federated Learning for Neural Networks: A Feature Learning Perspective

Yajie Bao, Michael Crawshaw, Mingrui Liu

International Conference on Machine Learning, 2024.

Federated Learning with Client Subsampling, Data Heterogeneity, and Unbounded Smoothness: A New Algorithm and Lower Bounds

Michael Crawshaw*, Yajie Bao*, Mingrui Liu (* denotes equal contribution)

Conference on Neural Information Processing Systems, 2023.

EPISODE: Episodic Gradient Clipping with Periodic Resampled Corrections for Federated Learning with Heterogeneous Data

Michael Crawshaw, Yajie Bao, Mingrui Liu

International Conference on Learning Representations, 2023.

Robustness to Unbounded Smoothness of Generalized SignSGD

(Alphabetical order) Michael Crawshaw, Mingrui Liu, Francesco Orabona, Wei Zhang, Zhenxun Zhuang

Neural Information Processing Systems, 2022.

Fast Composite Optimization and Statistical Recovery in Federated Learning

Yajie Bao, Michael Crawshaw, Mingrui Liu

International Conference on Machine Learning, 2022.

EMPLOYMENT

Flatiron Institute

May 2025 - August 2025

Summer Predoctoral Researcher, Center for Computational Mathematics

- Developed novel non-Euclidean gradient descent methods to train deep neural networks, especially language models.
- Proposed methods match or outperform state-of-the-art with significantly improved robustness to hyperparameter tuning.

Olive

March 2018 - August 2019

Machine Learning Engineering Intern

- Developed computer vision functionality for desktop automation software with applications to healthcare operations.
- Trained deep neural networks for object detection with various techniques, including Faster R-CNN and DARTS.

AWARDS

Top Reviewer , NeurIPS	2025
Top Reviewer , ICML	2025
Best Reviewer , AISTATS	2025
Top Reviewer , NeurIPS (top 8%)	2023
Institute for Digital Innovation Predoctoral Fellowship , George Mason University	2022
NSF XSEDE startup allocation , National Science Foundation	2020
Summer Ph.D. Research Initiation Award , George Mason University	2020
Outstanding Graduate Teaching Assistant , George Mason University	2020
Gordon Memorial Fund Scholarship , The Ohio State University	2017 - 2019
Honorable Mention, Raser-Bareis-Gordon Math Competition , The Ohio State University	2017
7th Place, FIRST Tech Challenge World Competition , FIRST	2015

ACADEMIC SERVICE

Reviewer , AAAI, ICLR	2026
Reviewer , ICLR, AISTATS (awarded Best Reviewer), ICML (awarded Top Reviewer), NeurIPS (awarded Top Reviewer)	2025
Reviewer , AISTATS, ICML, NeurIPS	2024
Reviewer , NeurIPS (awarded Top Reviewer, top 8%)	2023

TEACHING

Graduate Teaching Assistant , George Mason University	2019-2022
CS 657: Mining Massive Datasets	Fall 2020, Fall 2021
CS 471: Operating Systems	Fall 2020, Fall 2021, Spring 2022
CS 583: Analysis of Algorithms	Spring 2021
CS 571: Operating Systems	Spring 2021
CS 330: Formal Methods and Models	Fall 2019, Spring 2020
Undergraduate Teaching Assistant , The Ohio State University	2017 - 2018
CSE 3321: Automata and Formal Languages	Summer 2017, Fall 2017, Spring 2018
Undergraduate Honors Math Mentor , The Ohio State University	2016 - 2017
Math 4181H: Honors Analysis I	Fall 2016
Math 4182H: Honors Analysis II	Spring 2017

RELEVANT COURSEWORK AND SKILLS

GMU Coursework: Optimization for machine learning, deep learning, computer vision, theory of computation, algorithms, graphics, software testing.

OSU Math Coursework: Real analysis, linear algebra, differential equations, probability, statistics, combinatorics, complex analysis, number theory, abstract algebra.

OSU CS Coursework: Software, digital logic, databases, operating systems, networking, theory of computation, machine learning, neural networks, natural language processing, GPU programming (CUDA).

Programming: Python (PyTorch, TensorFlow), Bash, Java, C, Git, Latex